

Challenge (Habanero)

- Q1.** Evaluate the expression $3 \cdot 2 + 11 - 8$.
 (A) 6
 (B) 10
 (C) 12
 (D) 14
 (E) 16
- Q2.** Translate the phrase '5 more than twice a number' into an algebraic expression.
 (A) $2x + 1$
 (B) $2x + 5$
 (C) $2x - 5$
 (D) $x + 5$
 (E) $x - 5$
- Q3.** Simplify the algebraic expression $2x + 3x - x$.
 (A) $3x$
 (B) $4x$
 (C) $5x$
 (D) $6x$
 (E) x
- Q4.** Evaluate the expression $x^2 + 4x + 4$ when $x = 2$.
 (A) 12
 (B) 16
 (C) 20
 (D) 24
 (E) 28
- Q5.** A gamer has 120 minutes of gameplay left. If they use 15 minutes per level, how many levels can they complete? The expression to model this situation is $\frac{120}{15} - x$, where x is the number of levels completed. If $x = 2$, how many minutes are left?
- (A) 90
 (B) 80
 (C) 70
 (D) 60
 (E) 50
- Q6.** A coding app requires $3 \cdot 10^3$ lines of code. If a programmer codes $2 \cdot 10^2$ lines per day, how many days will it take to complete the code?
 (A) 10
 (B) 15
 (C) 20
 (D) 25
 (E) 30
- Q7.** The cost of data usage is 0.05 *per megabyte*. If a user uses $2.5 \cdot 10^3$ megabytes, what is the total cost?
 (A) 125
 (B) 130
 (C) 135
 (D) 140
 (E) 150
- Q8.** A scoreboard shows $5x + 2$ points for the home team and $3x - 1$ points for the visiting team. If $x = 4$, how many more points does the home team have?
 (A) 15
 (B) 20
 (C) 25
 (D) 30
 (E) 35

Open Ended Questions (FRQ)

- Q9.** A game developer uses the expression $2x^2 + 5x - 3$ to model the number of players in a game, where x is the number of hours since the game started. Evaluate the expression when $x = 2$ and explain the result in the context of the game.
- Q10.** A tech company uses the expression $3(2x + 1)$ to model the amount of data used by their servers, where x is the number of users. Simplify the expression and evaluate it when $x = 5$. Show all work and explain the result in the context of the company.

Chef's Tips

- Tip 1: Ensure students understand the order of operations (PEMDAS) to evaluate numerical expressions.
- Tip 2: Encourage students to use real-world examples, such as tech and gaming, to make algebra more engaging and relevant.
- Tip 3: Provide opportunities for students to practice simplifying algebraic expressions and evaluating them with given values.